

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-CT-7801-05 Sheet 1 of 1
OPL Title	Drift Eliminator & Fill Inspection	Discipline	Mechanical	
Equipment	CT-7801 - COOLING TOWER CELL FAN	Area / Unit	7800 - COOLING TOWER SYSTEM	
Related Interlock	SEQ-7801 (SIL 1)	P&ID; Ref	TJC-LLD-PID-7801	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **drift eliminator & fill inspection** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the PVC fill for scaling and collapse	Isolation & LOTO verified
2	Check the drift eliminator packs are aligned	Parts match BOM / spec
3	Hose down fouled sections during a cell outage	Torque / clearance to spec
4	Report a thermal performance drop via TT-7801	No leak / smooth operation
5	Plan fill replacement if degradation is widespread	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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